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Computer Science & IT

Computer Networks

Transport Layer

DPP: 1

Q1 Consider total length field value of UDP packet header is 0x005C (in hexadecimal), calculate its payload size (in bytes) ?

Q2 Which transport layer protocol is used for faster communication ?
(A) TCP
(B) UDP
(C) IP
(D) None of the above

Q3 Consider a TCP sender sends two segments of same size of same session to TCP receiver one after another. Both the segments carry equal size data and received successfully by receiver, sequence number field values in headers (in decimal) are 1500 and 1800 respectively. Receiver sends individual acknowledgment for each segment, what are acknowledgement numbers send by receiver ?
(A) 1500 and 1800 (B) 1799 and 2099
(C) 1800 and 2100 (D) 2100 and 2100

Q4 Consider the three-way handshake for TCP connection establishment. Identify TCP state(s) for which TCP client transit ?
(A) LISTEN (B) SYN_SENT
(C) SYN_RECV (D) ESTABLISHED

Q5 Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. The TCP server send ACK to acknowledges the FIN segment, in which state does the server side TCP connection wait to close the connection ?
(A) CLOSE-WAIT (B) TIME-WAIT
(C) FIN-WAIT-1 (D) FIN-WAIT-2

Q6 Which socket API functions executed by TCP client?
(A) bind() (B) connect()
(C) listen() (D) accept()

Q7 Consider size of congestion window of a TCP connection be 8 MSS when a timeout occurs. The round trip time of the connection is 2 seconds. The time taken (in sec) by the TCP connection to reach upto 9 MSS congestion window is _____.

Q8 Consider TCP protocol runs over a 100 Mbps network. What should be minimum time (in seconds, rounded off to nearest integer) required for maximum segment lifetime?

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Answer Key

Q1	84	Q5	(A)
Q2	(B)	Q6	(B)
Q3	(C)	Q7	14
Q4	(B, D)	Q8	344

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Hints & Solutions

Q1 Text Solution:

Total length = 0 × 005C = (5 × 16 + 12)

= 92 bytes

Payload Size = Total length – Header size

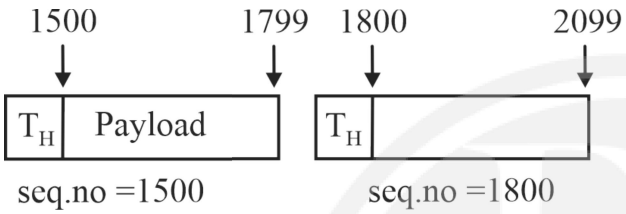
= 92 bytes – 8bytes

= 84 bytes

Q2 Text Solution:

(Basic Protocol)

Q3 Text Solution:



Size of payload = (1800 – 1500) byte

= 300 bytes

ACK no. = (1500 + 300) and (1800 + 300)

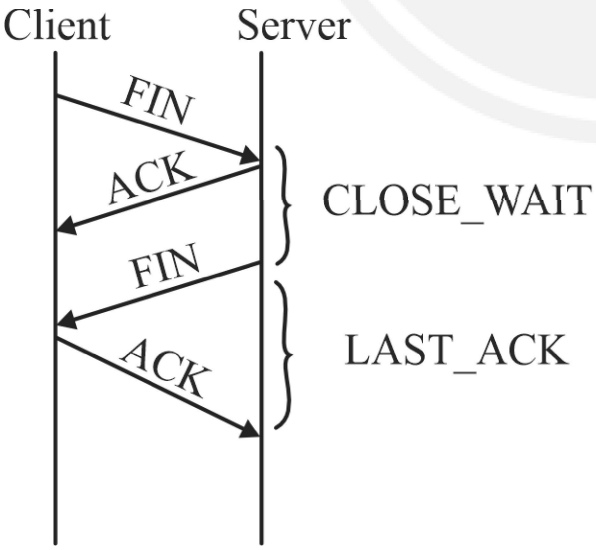
= 1800 and 2100.

Q4 Text Solution:

TCP Server

(B & D)

Q5 Text Solution:



Q6 Text Solution:

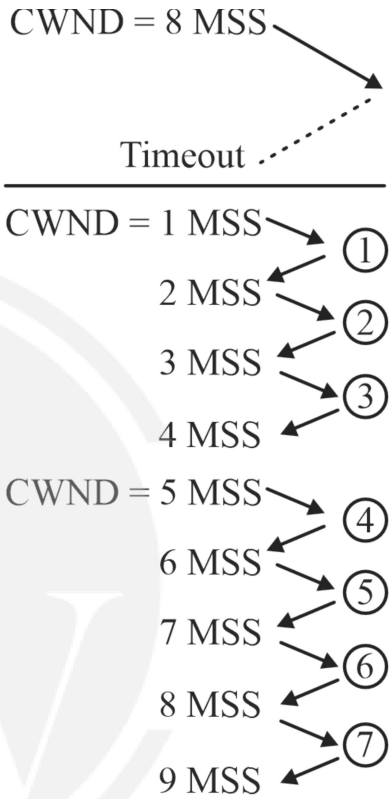
TCP Server

TCP Client

TCP Server

TCP Server

Q7 Text Solution:



SSThresh = $\frac{\text{CWND}}{2}$ $\frac{\text{CWND}}{2}$

= $\frac{8\text{ MSS}}{2}$ $\frac{8\text{ MSS}}{2}$

= 4 MSS

= 7 × RTT

= 7 × 2 sec

= 14 sec.

Q8 Text Solution:

Min^m value for MSL = $\frac{2^{32}\text{ bytes}}{\text{Bandwidth}} = \frac{2^{35}\text{ bits}}{100\text{ Mbps}}$

$\frac{2^{32}\text{ bytes}}{\text{Bandwidth}} = \frac{2^{35}\text{ bits}}{100\text{ Mbps}}$

= $\frac{2^{35}\text{ bits}}{10^8\text{ bits / sec}} = 343.59\text{ sec}$

$\frac{2^{35}\text{ bits}}{10^8\text{ bits / sec}} = 343.59\text{ sec}$

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= 344 sec

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